permissions for executing a command similar to another user, for instance, the superuser. Sudo gives authorised users the provision to execute commands as another user, - specified in the / etc / sudoers file. The effective and real GID and UID of the issuing user are matched with those of the target user as laid down in the password file. By default, sudo requires users to authenticate themselves using a password. This authentication involves the user specific password and not the root password itself.

After user authentication, a timestamp is logged, and the user can use sudo for a short period without a password. This timestamp can be updated if the user issues sudo command using the -v tag. If an unauthorised user tries to execute a command via sudo, the attempt to breach system security will be considered unsuccessful, and an email will be sent to the appropriate authority. The default permission given for failing to execute this command lies with the root. Note that if an unauthorised user tries to run sudo using the -l or -v tags, no mail will be sent. This is so because these commands enable users to determine if they are allowed to use the sudo command.

Sudo can record successful and failed attempts (in addition to errors) in the system log(syslog) , a unique log file, or both. By default, sudo makes the entry into the syslog, but this can be changed during configuration time or in the sudoers file. If the execution of the program is successful, the return value from sudo will be the return value of the program executed.

Otherwise, sudo exits a port value of 1 if there is a configuration/authorization problem or if it cannot execute the specified command. In the latter case, the error string is exported to 'stderr'. If sudo cannot enter one or more entries in the user's PATH, an error is printed to stderr. This usually does not occur under normal circumstances. The most common reason for returning the "permission denied" is when you make an automatic transition and one of your PATH directories is on a device that is currently unreachable.

Sudo strives to be safe when commands are executed. Variables that determine the mannerism of dynamic loading and binding can be utilised for subverting a program that is being run by sudo. As a resolution, it may remove some environmental variables specific to the system, particularly from the environment that issues the commands which are to be executed. Examples:

- `sudo shutdown -r now` (This command will restart the system and execute the shutdown command as root)
- `sudo -u Mary ls /home/David/Documents` (This command will list g the content of the home/David/Documents directory as the user Mary)
- `sudo -u Mary -g David mkdir /home/otheruser/Documents/newfiles`